Adult Income Prediction: Problem Framing & Baselines

1 PROBLEM FRAMING & METRIC

Goal: Predict whether a person earns > $50K/year to prioritize outreach (e.g., for premium products) toward likely qualified prospects. Positive class: income > $50K. Base rate: **23.9%** of individuals earn > $50K (11,687 of 48,842).

Primary metric: Precision (minimizes costly false positives), with **ROC AUC** as the secondary, threshold-free metric. Precision is interpreted alongside recall to ensure the model maintains useful coverage rather than achieving high precision by predicting only a few positives.

The hold-out test set is touched once for final reporting; using it during tuning would inflate reported performance.

model	accuracy	precision	recall	f1	roc_auc	pr_auc
Majority-class	0.7607	0	0	0	0.5	0.2393
Rule-based	0.7520	0.4852	0.5967	0.5352	0.7493	0.5079

3 BASE RESULT ON HOLD OUT DATA

The majority-class baseline reaches **76.1% accuracy** but **0 precision/recall** (it never predicts >50K), showing why accuracy is misleading on this imbalanced dataset. The rule-based baseline (education-num >= 13 OR capital-gain > 0) achieves **Precision = 0.4852**, **F1 = 0.5352**, and **ROC AUC = 0.7493**. Since this project is evaluated primarily on precision, the benchmark to beat is **Precision = 0.4852**, with ROC AUC used as the secondary comparison.

4 ERROR ANALYSIS & NEXT STEPS

False negatives (missed high earners) skew toward married individuals in executive/professional occupations with high hours-per-week but no capital-gain and sub-Bachelor's education, which the rule ignores. False positives skew toward younger, highly educated people early in their careers who have not yet reached >$50K. Missing workclass/occupation values appear in both error groups.

Primary metric going forward: Precision (secondary: ROC AUC). **Baseline to beat:** Precision = 0.4852, ROC AUC = 0.7493.

Fixes:

1. Handle missing workclass/occupation/native-country as an explicit Unknown category.

2. One-hot encode categorical variables.

3. Drop the redundant string education column (keep education-num).

4. Address skew in capital-gain/capital-loss (log transform or binarize).

5. Test with/without fnlwgt (a sampling weight rather than a predictive feature).

6. Bucket rare native-country values into Other.